

Modern Two-Linear-Forms Möbius and Liouville Correlation Estimates: A Comprehensive Landscape Map

The study of multiplicative functions, specifically the Möbius function $\mu(n)$ and the Liouville function $\lambda(n)$, lies at the heart of analytic number theory. A central heuristic in this domain asserts that the sign patterns of these functions behave like a sequence of independent, completely random variables taking values in $\{-1, 1\}$ ¹. The primary obstacle to proving this pseudorandomness is the estimation of joint correlations along linear forms⁴. The most famous formulation of this decoupling is the Chowla conjecture, which in its two-point (or two-linear-forms) guise predicts the asymptotic orthogonality of shifted copies of the Liouville function⁴.

Over the last decade, a series of profound breakthroughs has radically transformed the landscape of these estimates. By integrating techniques from information theory, additive combinatorics, expander graphs, and algebraic geometry, mathematicians have established rigorous boundaries for these correlations across both the ring of integers $\mathbb{Z}$ and the polynomial ring over a finite field $\mathbb{F}_q[T]$ ⁶.

1. Sieve-Theoretic Obstructions and the Parity Barrier

To establish the context of modern two-point correlation estimates, one must first confront the classical parity barrier of sieve theory⁴. The two-point correlation of the Liouville function is defined for distinct shifts $h_1, h_2 \in \mathbb{Z}$ as the sum:

$$C(X; h_1, h_2) = \sum_{n \leq X} \lambda(n + h_1) \lambda(n + h_2)$$

The $k = 1$ case of this correlation is equivalent to the Prime Number Theorem³. For $k \geq 2$, however, the problem becomes notoriously difficult and is closely related to the Twin Prime Conjecture⁴.

Traditional sieve methods, which are designed to estimate the density of prime tuples, fail to control the behavior of $\lambda(n)$ because of the parity problem⁴. Combinatorial and Selberg-type sieves are structurally unable to distinguish integers with an odd number of prime factors from

those with an even number of prime factors⁴. Since the Liouville function is defined precisely as $\lambda(n) = (-1)^{\Omega(n)}$, where $\Omega(n)$ counts the total number of prime factors with multiplicity,

classical sieves cannot produce non-trivial upper bounds on $C(X; h_1, h_2)$ without additional structural inputs⁴.

The connection between two-point correlations and the global behavior of the Liouville function is formalized via the classical Square Identity¹:

$$|L(X)|^2 = X + 2 \sum_{1 \leq h < X} C_h(X - h)$$

where $L(X) = \sum_{n \leq X} \lambda(n)$, and $C_h(Y) = \sum_{n \leq Y} \lambda(n)\lambda(n + h)$ represents the fixed-endpoint correlation¹. To bound the summatory Liouville function at the square-root scale $L(X) \ll X^{1/2+\epsilon}$, which is classically equivalent to the Riemann Hypothesis, one must control the maximal correlation energy across dyadic blocks¹.

Defining the collar energy over a dyadic block H as:

$$E(X, H) = \sum_{H < h \leq 2H} C_h^*(X)^2$$

where $C_h^*(X) = \sup_{1 \leq Y \leq X-h} |C_h(Y)|$ represents the moving-endpoint correlation, the Riemann Hypothesis is recovered if the collar energy satisfies the uniform bound¹:

$$E(X, H) \ll_{\epsilon, \eta} X^{2+\epsilon} H^{-1}$$

for all dyadic $1 \leq H \leq X^{1-\eta}$ ¹. This formulation reveals that the moving-endpoint supremum introduces a significant technical hurdle¹.

By exploiting endpoint duality, the maximal correlation energy can be expressed as a dual norm over the unit sphere of moving endpoints¹. This duality allows one to upgrade fixed-endpoint spectral large sieve bounds to their maximal-endpoint forms with no power loss, meaning that the moving endpoint is not the fundamental bottleneck in proving the Riemann Hypothesis¹.

Instead, the bottleneck is coefficient-uniform failure¹. For any real scale X , the collar-scale

bound for the sum $\sum_{n \leq Y} \lambda(n)\lambda(n+h)$ cannot follow from the boundedness of the coefficients alone¹. There exist explicit bounded sequences that force the corresponding correlation energy above the required scale by a fixed power of X , demonstrating that the multiplicative sign structure of $\lambda(n)$ is the fundamental source of correlation decay, rather than generic analytic properties of the sum¹.

2. Integer Landscape: Logarithmic Averaging and Scale Propagation

Because of the sieve-theoretic parity barrier, progress on the integer ring $\mathbb{Z}$ requires working with averaged or modulated formulations of the correlation sums⁶.

Typical Short-Interval Dynamics

A major breakthrough occurred when Kaisa Matomäki and Maksym Radziwiłł established that bounded multiplicative functions exhibit significant cancellation on typical short intervals

$[x, x+H]$, provided that the interval length $H(X)$ tends to infinity with X ¹².

Unconditionally, for the Liouville function, they proved¹³:

$$\lim_{X \rightarrow \infty} \frac{1}{X} \int_X^{2X} \left| \frac{1}{H} \sum_{x < n \leq x+H} \lambda(n) \right|^2 dx = 0$$

This results from a deep analysis of the Dirichlet polynomial of $\lambda(n)$ near the 1 -line, showing

that the function cannot "pretend" to be a twist n^{it} for large t on average over short intervals¹⁵. This short-interval cancellation represents a complete resolution of the

typical-interval problem, as H can grow arbitrarily slowly, bypassing the classical zero-density barriers such as $H \geq X^{7/12}$ ¹³.

The Logarithmically Averaged Two-Point Estimate

By building upon the Matomäki-Radziwiłł theorem, Terence Tao established the first non-trivial case of the two-point Chowla conjecture⁶. Tao introduced a logarithmic average to establish the estimate⁶:

$$\sum_{X/\omega \leq n \leq X} \frac{\lambda(a_1 n + b_1) \lambda(a_2 n + b_2)}{n} = o(\log \omega) \quad \text{as } \omega \rightarrow \infty$$

for any natural numbers a_1, a_2 and integers b_1, b_2 such that $a_1 b_2 - a_2 b_1 \neq 0$.⁶ The primary mechanism that makes logarithmic averaging tractable is dilation invariance⁶. The logarithmic measure dn/n is approximately invariant under the scaling map $n \mapsto pn$ for any prime p . If the correlation sum was large and positive, the multiplicativity of the Liouville function ($\lambda(pn) = \lambda(p)\lambda(n)$) would imply that the correlation must remain large and positive when restricted to arithmetic progressions modulo p for a vast set of medium-sized primes P .

The Entropy Decrement Argument

To prove that this correlation must in fact vanish, Tao designed the *entropy decrement argument*⁷. The proof reformulates the sequence of signs $(\lambda(n+1), \dots, \lambda(n+H))$ as a random variable on the hypercube $\{-1, 1\}^H$, where the index n is chosen uniformly at random⁶. One then constructs a bipartite graph $G_{n,H}$ on the vertex set $\{1, \dots, H\}$, where an edge exists between i and j if there is a prime $p \in P$ dividing $n+i$ and $n+j$. If the shifted values of λ correlated strongly, the random sequence of signs would share a substantial amount of mutual information with the divisor graph $G_{n,H}$. However, by analyzing the Shannon entropy of the joint distributions, one observes that the total entropy of the system is bounded by the topological dimension of the hypercube⁶. Using the subadditivity of the entropy function, Tao proved that the mutual information cannot grow indefinitely⁶. By systematically increasing the scale parameter H across a dyadic range, one must reach a metastable region where the mutual information shared between the sign sequence and the graph is negligible⁶. In this state of asymptotic independence, the correlation sums are forced to collapse, yielding the desired cancellation⁶.

In establishing this concentration, the uniform random variable Y_H over the interval must behave consistently when conditioned on the divisor states²¹. This requires analyzing the

concentration of measure for the conditional variables $Y_H|X_H = x$ for each residue class, which is achieved via refinements of Hoeffding's inequality²¹.

Quantitative Parameter	Entropy Decrement (Tao)	Expander Graphs (Helfgott)
Logarithmic Decay Rate	$o(\log \omega)$	$O\left(\frac{1}{\sqrt{\log \log X}}\right)$
Underlying Graph Structure	Random Bipartite Divisor Graph $G_{n,H}$ [cite: 6]	1-Sided Expander $A _X$ [cite: 9, 22]
Spectral Gap Property	Mutual Information Minimization ⁶	Eigenvalue Bound: $\rho \ll \sqrt{L}$ [cite: 22]
Walk Model	Multiplicativity at Small Primes ⁶	Random Walks of Length $k = 2\ell$ [cite: 9]

3. Alternative Integer Paradigms: Expanders, Walks, and Siegel Zeroes

The soft bounds generated by the entropy decrement argument have motivated the development of alternative, quantitatively superior paradigms over $\mathbb{Z}_9$.

Expander Graphs on Divisibility Chains

Harald Helfgott and Maksym Radziwiłł developed an expander-graph framework that yields sharp, power-saving decay rates⁹. Instead of relying on mutual information depletion, they proved that a linear operator A defined in terms of prime divisibility over a typical subset $X \subset \mathbb{N}$ has a strong expander property⁹. Specifically, for a prime set P of cardinality L , they showed that the eigenvalues of the adjacency matrix restricted to X satisfy the bound $\rho \ll \sqrt{L}$.

This expander property implies that for any bounded sequences $f, g : \mathbb{N} \rightarrow \mathbb{C}$, the

correlation over divisibility steps satisfies²²:

$$\frac{1}{NL} \sum_{n \in N} \sum_{p \in P, p|n} f(n) \bar{g}(n \pm p) - \sum_{n \in N} \sum_{p \in P} \frac{f(n) \bar{g}(n \pm p)}{p} = O\left(\frac{1}{\sqrt{L}}\right)$$

By specializing this bound to the Liouville function ($f(n) = g(n) = \lambda(n)$) and utilizing the Matomäki-Radziwiłł-Tao theorem, Helfgott obtained a sharp decay rate of $O(1/\sqrt{\log \log X})$ on the logarithmic scale, which represents a massive quantitative improvement over the entropy decrement bounds⁹.

To analyze the propagation of the signs of λ along these expander graphs, Helfgott modeled the divisibility chains as a random walk of length $k = 2\ell$. The probability of a walk consisting of steps $p_1, \dots, p_k$ being allowed is shown to be asymptotically $N/(p_1 \cdots p_k)$. By approximating the distribution of the endpoint of the walk using a continuous function $\psi(m)$, the correlation is bounded by the double sum⁹:

$$\left| \sum_{n \in N} \sum_m \lambda(n) \lambda(n + m) \psi(m) \right| \leq \text{err}_2 \cdot NL^k$$

Evaluating this error via the Matomäki-Radziwiłł theorem demonstrates that the sign correlations decay at a rate consistent with a random walk on a high-dimensional torus⁹.

Non-Backtracking Walks and Local Fourier Uniformity

The landscape of modulated two-point estimates has been further extended by Cédric Piltat, who investigated the Fourier Uniformity Conjecture¹⁰. The conjecture predicts that the supremum of the correlation of the Liouville function with any linear phase $e(n\alpha)$ on almost all short intervals of length H decays to zero¹⁰.

The primary barrier to establishing this conjecture was the threshold

$H \approx \exp((\log X)^{1/2})$ ¹⁰. Standard iterative arguments along divisibility graphs lose a

factor of $k!$ at step $k \approx \frac{\log X}{\log H}$, which becomes catastrophic below this scale¹⁰.

By employing non-backtracking walks on the divisor graphs, Piltat eliminated the

self-intersection of paths during the scale-propagation phase¹⁰. This technique removes the $k!$ losses, yielding unconditional cancellation for interval lengths as short as $H \geq \exp((\log X)^{2/5+\epsilon})$ ¹⁰.

The Impact of Siegel Zeroes

An intriguing feature of the modern landscape is the behavior of these correlation estimates in the presence of a Siegel zero²⁵. If a Siegel zero β exists with associated conductor q_χ and quality η , the normal statistical behavior of both the von Mangoldt function $\Lambda(n)$ and the Liouville function is disrupted over vast scales²⁵. Tao and Teräväinen established a hybrid version of the Chowla and Hardy-Littlewood conjectures under this hypothesis²⁵. They proved that for an infinite sequence of scales x satisfying $q_\chi^{10k+1/2+\epsilon_0} \leq x \leq$, the hybrid correlation of k copies of Λ and ℓ copies of λ satisfies²⁶:

$$\sum_{n \leq x} \Lambda(n + h_1) \cdots \Lambda(n + h_k) \lambda(n + h'_1) \cdots \lambda(n + h'_\ell) = \mathcal{S} + O\left(\frac{1}{\log^{1/10} \eta}\right)$$

where $\mathcal{S}$ is the singular series associated with the prime tuple²⁶. If $\ell \geq 1$, the singular series $\mathcal{S}$ vanishes, confirming the Chowla conjecture unconditionally within this range²⁵. This hybrid formula unifies several prior conditional results, which are summarized below:

Hybrid Context	Siegel Zero Range (x)	Asymptotic Correlation Estimate	Source Attribution
Twin Primes (Λ, Λ)	$q_\chi^{250} \leq x \leq$	$\mathcal{S} +$	Heath-Brown ²⁵
Möbius Pair (λ, λ)	$q_\chi^{10} \leq x \leq$	$O\left(\frac{1}{\log \log \eta}\right)$	Germán-Kátai ²⁶

k -Point Chowla	$q_\chi^{10} \leq x \leq$	$O\left(\frac{1}{(\log \log \eta)^{1/2} \log^{1/12}}\right)$	Chinis ²⁵
Hardy-Littlewood-Chowla	$q_\chi^{10k+1/2+\epsilon_0} \leq x \leq$	$\mathcal{S} +$	Tao-Teräväinen ²⁵

4. Function Field Paradigm: Geometrization and Algebraic Duality

In the global function field setting over a polynomial ring $\mathbb{F}_q[T]$, the landscape of modern two-linear-forms estimates is governed by powerful geometric techniques that dissolve the classical parity barrier²⁷.

Large- q Limits and Discriminant Square Independence

For any finite field $\mathbb{F}_q$ of odd characteristic, the Möbius function $\mu(F)$ of a monic polynomial $F \in \mathbb{F}_q[T]$ of degree n is linked to the quadratic character χ_2 of its algebraic discriminant³⁰:

$$\mu(F) = (-1)^{\deg F} \chi_2(\text{disc}(F))$$

For distinct shift polynomials $\alpha_1, \dots, \alpha_r$ of degree $< n$, the correlation sum over the set M_n of monic polynomials of degree n is:

$$C(\alpha_1, \dots, \alpha_r; n) = \sum_{F \in M_n} \chi_2 \left(\prod_{i=1}^r \text{disc}(F + \alpha_i) \right)$$

By parameterizing the polynomial as $F(x) = f(x) + t$, where $t \in \mathbb{F}_q$ is the constant term and $f(x)$ represents the trace-free component, the inner sum over t becomes a one-dimensional character sum of the high-degree polynomial

$$P(t) = \prod_{i=1}^r \text{disc}(f(x) + t + \alpha_i) \quad {}_{30}.$$

Carmon and Rudnick proved that for a dominant portion of "good" coefficients

$f(x) \in \mathbb{F}_q^{n-1}$, the discriminants $\text{disc}(f(x) + t + \alpha_i)$ are pairwise coprime and square-free³⁰. Applying the Hasse-Weil bound to the resulting hyperelliptic curves yields the large- q estimate³⁰:

$$|C(\alpha_1, \dots, \alpha_r; n)| \leq rnq^{n-1/2} + \frac{3}{4}(r+3)n^2q^{n-1}$$

which establishes that the correlations vanish as $q \rightarrow \infty$ ³⁰.

In characteristic 2, the classical discriminant of any polynomial is a perfect square, carrying no information about the parity of its irreducible factors³². To resolve the Chowla conjecture in even characteristic, Dan Carmon utilized Berlekamp's discriminant³³. For a polynomial F with roots $r_1, \dots, r_n$ in the algebraic closure, Berlekamp's discriminant is defined as³²:

$$\text{Berl}(F) = \sum_{i < j} \frac{r_i r_j}{r_i^2 + r_j^2}$$

The corresponding character relation in characteristic 2 is given by

$\mu(F) = (-1)^{\deg F} \chi_2(\text{Berl}(F))$, where $\chi_2(x) = (-1)^{\text{Tr}(x)}$ is the additive quadratic character³². By proving square independence for Berlekamp discriminants via resultants and lifting them to characteristic zero, Carmon recovered the identical bound of $O_r(q^{n-1/2})$ over fields of even characteristic³³.

Fixed- q , Large-Degree Limits and Sheaf Cohomology

A much deeper challenge is the fixed- q , large-degree limit ($n \rightarrow \infty$), which is structurally analogous to the integer setting²⁷. This was resolved in a landmark paper by Will Sawin and

Mark Shusterman, who confirmed the k -point Chowla conjecture over $\mathbb{F}_q[T]$ for any odd prime characteristic satisfying $q > p^2 k^2 e^2$ ⁸.

Their strategy exploits a remarkable algebraic property of positive characteristic: for any fixed

polynomial r , the Möbius function evaluated along the progression $r + sp^d$ is given exactly by a quadratic Dirichlet character⁸:

$$\mu(r + sp^d) = \chi_{D_r}(s + c_r)$$

This allows the short correlation sum to be written as a short sum of additively shifted Dirichlet characters⁸.

To estimate these sums, Sawin and Shusterman expressed the problem geometrically, mapping

the short interval to an affine space and modeling the character χ_{D_r} as an ℓ -adic sheaf⁸. By applying Deligne's proof of the Weil conjectures and the theory of vanishing cycles, they analyzed the cohomology of this sheaf⁸. They proved that vanishing cycles can only occur when the singular locus of the character's conductor fails to form a simple normal crossings

divisor⁸. This geometric control yielded a level of distribution close to 1 for the Möbius

function, successfully resolving the k -point Chowla conjecture⁸.

These geometric techniques have also been applied to study the variance of arithmetic functions in short intervals³⁷. Ofir Gorodetsky obtained asymptotic formulas for the variance of square-free polynomials and factorization functions over short intervals, demonstrating that sums over random short intervals enjoy square-root cancellation³⁹. Furthermore, Keating and

Rudnick computed the variance of the von Mangoldt function $\Lambda(f)$ over short intervals $I(A; h)$ of size $h \leq n - 4$ ³⁷.

$$\lim_{q \rightarrow \infty} \frac{1}{q^{h+1}} \left(\frac{1}{q^n} \sum_{f \in M_n} |C_\Psi(f; h)|^2 \right) = n - h - 2$$

which represents the function-field analogue of the Goldston-Montgomery conjecture³⁷. This variance estimate was subsequently generalized to the covariance of counts of almost-primes

using higher-order von Mangoldt functions Λ_j , where the limiting distribution is controlled by the Haar measure of the unitary group $U(n - 1)$ via Katz's equidistribution theorem³⁷.

5. Modern Formalization: Verifying Correlation Infrastructure in Lean 4

The conceptual complexity of these modern proofs has driven an active effort to formalize additive combinatorics and analytic number theory using the Lean 4 interactive theorem prover and its mathematical library, Mathlib⁴².

Formalizing the Entropic Freiman-Ruzsa Theorem

The mathematical foundation of Tao's entropy decrement argument—specifically the entropic Ruzsa calculus—was formalized as part of the collaborative project to verify the Polynomial Freiman-Ruzsa (PFR) conjecture⁴⁵. The project formalized the entropic version of Marton's conjecture over $\mathbb{F}_2^n$ ⁴⁶. The entropic Ruzsa distance between two random variables X and Y is defined as⁴⁹:

$$d(X, Y) = \mathbb{H}(X' - Y') - \frac{\mathbb{H}(X') + \mathbb{H}(Y')}{2}$$

where X', Y' are independent copies of X, Y ⁴⁹. The formalization successfully proved the distance decrement property⁴⁹:

$$d(X_1 + X_2, Y_1 + Y_2) \leq d(X_1, Y_1) + d(X_2, Y_2) - c \cdot \Delta_d$$

and the crucial fibring inequality, which bounds the entropy of a joint distribution in terms of its image under a homomorphism and the average entropy of its fibers⁵⁰.

The formalization project also contrasted different design decisions in modeling information theory⁵². While the core project introduced a specialized `ProbabilitySpace` class, Rémy Degenne developed a mathlib-only approach that avoids type-theoretic bottlenecks⁵². This implementation defines entropy directly on top of Mathlib's measure-theoretic structures,

representing the law of a random variable X as the image measure `volume.map X` and utilizing standard `ENNReal` coercions to handle the real-valued Shannon entropy

$$\mathbb{H}(X) = - \sum p_i \log p_i \quad ^{52}.$$

Verification of Analytic and Geometric Foundations

Beyond additive combinatorics, key components of analytic number theory and algebraic geometry have been formalized in Lean 4 to support future correlation proofs⁵⁴:

- **The Prime Number Theorem:** The "PrimeNumberTheoremAnd" project formalized the asymptotic behavior of the prime counting function $\psi(x)$ with an explicit error term, which represents the necessary baseline for $k = 1$ correlation estimates⁵⁴.
- **Dirichlet L -Functions:** Dirichlet characters and their associated L -series have been

formalized in Mathlib, including Dirichlet's theorem on primes in arithmetic progressions and a formal statement of the Riemann Hypothesis⁵⁵.

- **The Mason-Stothers Theorem:** The polynomial version of the abc conjecture was formalized in Lean 4, proving that for coprime polynomials $a, b, c \in k[T]$ satisfying $a + b + c = 0$, the maximum degree is bounded by the degree of their radical⁵⁶. This theorem provides the algebraic foundation for analyzing polynomial curves over function fields⁵⁷.
- **The Schwartz-Zippel Lemma:** Formalized under `MvPolynomial.schwartz_zippel_totalDegree`, this lemma bounds the probability that a multivariate polynomial evaluates to zero over a finite grid, which is a key tool for certifying the algebraic independence of polynomial discriminants⁵⁸.
- **The C1 Spectroscope Mechanism:** Formalized as part of a verified bridge between modular forms and prime distributions, this mechanism integrates Hecke convolution operators T_h , holomorphic cusp form projectors P_{holo} , and regularized resolvent kernels $R_h(s)$ to analyze prime fractions over Farey arcs⁵⁹. The formalization verified the synthesis identity showing that the signed contribution of the double parabolic cross term lies within a tight spectral cage, reducing a ratios-conjecture computation to a single Eisenstein-side residue⁵⁹.

This extensive formalization network ensures that the highly technical, multi-disciplinary proofs characterizing modern two-point correlation estimates are verified to their logical foundations, providing a certifiably correct map of the landscape⁶⁰.

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